methionine. Similarly, phenylalanine can be converted to tyrosine, but not vice-versa. yet these spare the requirements of the corresponding essential amino acid. Hence, cystine and tyrosine are sub-classed as semi-essential amino acids.

III. Non-Essential Amino acids (Dispensable)

These amino acids can be synthesized in the body and not necessarily obtained through dietary intake.

The nutritional classification of amino acids is presented in **table 10.1**

10.3.2 Classification of Proteins (Based on chemical composition)

a) **Simple proteins**: It is composed entirely of amino acids only.

b) **Conjugated or Complex proteins:** It is made up of amino acids and other organic or inorganic compounds.

Table 10.1: Nutritional classification of amino acids

Essential	Semi-essential	Non-essential
Histidine	Arginine	Glutamic acid
Lysine	Tyrosine	Aspartic acid
Tryptophan	Cystine	Alanine
Phenylalanine	Glycine	Proline
Methionine	Serine	Hydroxyproline
Threonine		Cysteine
Leucine		
Isoleucine		
Valine		

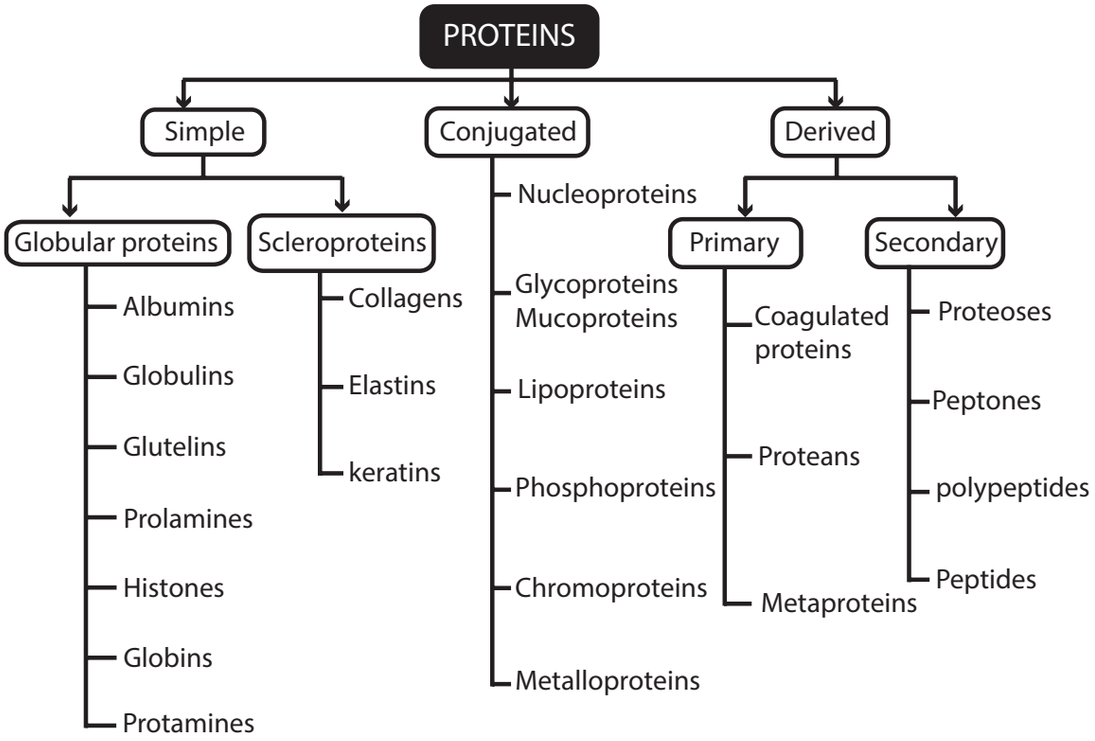


Fig 10.3: Classification of Proteins based on chemical composition

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